

Scale Navigator: a System for Networked Algorithmic Harmony

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ABSTRACT

We present the Scale Navigator, a graphical interface for navigating pandiatonic harmonies and harmonically synchronizing performers in a laptop orchestra environment. We describe its use both as a composition tool—in which a user manipulates the tool’s web interface to generate chord progressions—and in live performance, in which a conductor explores the Scale Navigator’s network of tonal scales and broadcasts scalar control data, MIDI and real-time notation to an ensemble of live electronic performers, sight-reading improvisers playing acoustic instruments, and musical generative algorithms.

Author Keywords

NIME, networked music, performance, composition, live score, improvisation, notation, sight-reading, interaction, scale network, pandiatonicism, harmony, JavaScript, Node, P5, VexFlow, WebSockets, Python

CCS Concepts

•Applied computing → Sound and music computing; Performing arts; •Human-centered computing → Graphical user interfaces;

1. INTRODUCTION

The Scale Navigator is an implementation of Dimitri Tymoczko’s scale network graph, in which each node is a different tonal 7-note scale. Edges are drawn between nodes that share “maximally intersecting voice leading,” which Tymoczko describes as “voice leading in which only a single pitch class moves, and it moves by only a single semitone.” The resulting network—after implementation with JavaScript and P5.js—is an interface that allows the user to navigate from scale to scale. Each new movement to an adjacent node generates notation and MIDI control data that describes the selected scale. Notation is rendered via VexFlow in the upper left corner of the interface for improvisers to

sight-read and interpret. Control data is routed to Ableton Live(or similar DAW) as MIDI to manage various parameters, mainly the scale object that determines which MIDI pitch classes are ‘allowable.’ The interface also generates progressions of jazz chords and outputs these to a MIDI channel separate from that of the control data.

The Scale Navigator becomes the Conductor module of a system for networked music performance when connected to a multi-nodal server. This server uses WebSockets to mediate messages from the Scale Navigator to Client pages open in the performers’ web browsers, which displays notation describing the current scale as selected by the Conductor. The Client page also has a button for mapping various streams of incoming control data to parameters in Ableton Live.

From an aesthetic standpoint, the Scale Navigator arose from a desire to explore post-tonal diatonicism within the framework of Tymoczko’s scale network concept, particularly music with a Pandiatonic harmonic language as described by Slonimsky and used by Richard Grayson. Tymoczko’s framework is flexible enough to support not only Debussy’s harmonic language but also other more extreme examples of non-functional approaches to diatonic harmony, such as the combinatorial / permutative language of Terry Riley’s *In C* and the diatonic set-theory method Santa’s *Analysing Post-Tonal Diatonic Music: A Modulo 7 Perspective*.

2. DESIGN

The Scale Navigator is flexible and multimodal in its uses. 1) As a composing tool, it affords users a framework for modulation between maximally-intersecting key areas, and generates smoothly voice-led jazz chord progressions as subsets of these keys. 2) In a live performance setting, the interface becomes an instrument for improvisation, sending control data and MIDI to a DAW such as Ableton Live for further on-the-fly manipulation and sound design. 3) The Scale Navigator’s functionality expands further when using WebSockets to send these messages over a server to fellow performers, and it becomes an interface for a conductor to harmonically synchronize players in an electro-acoustic ensemble.

2.1 Scale Navigator / Conductor

On its surface, the Scale Navigator is a composing tool for deciding which key areas to modulate to. Each node represents a 7-note Pressing scale, and has 4 or 6 other scale-nodes that share maximally smooth voice leading—this pat-



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NIME’79, June 3-6, 2019, Porto Alegre, Brazil.

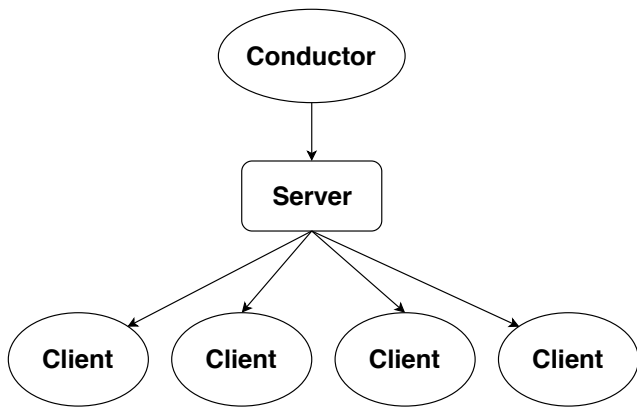


Figure 1: Signal Flow of the Scale Navigator Networked Harmony Environment

tern determines the relationship between nodes. In an attempt Colors of these nodes are determined by root movements by fourths are along the y axis discuss colors, shapes and placement. renders notation of the 7 notes of the scale in upper left corner. UI design for scale nav—other thoughts?

as a composing too, generates chord progressions: discuss algorithms for generating these. 1) dataset of Jazz Chords found in Bill Boyd's Jazz Chord Progressions. Nathan Ho implements algorithm of Boyd's for picking the NEXT ONE: chord voicings / positions are determined by root movement. Nathan Ho introduces a voice leading algorithm to rank the list of possible next chords according to a fitness function: smooth voice leading for a higher score / higher ranking. all of this fits within the context of the scale network.

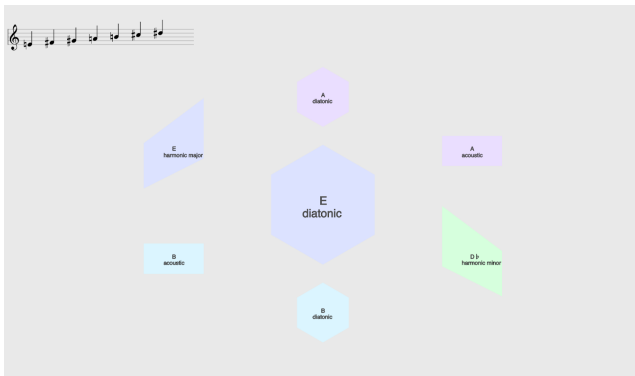


Figure 2: Scale Navigator user interface

2.2 Client

The Client page is purely functional, and is intended for use by both sight-readers and electronic improvisers. The notation and scale label are rendered in the upper left hand corner

2.3 Server

talk to colin and dexter about this???? maybe doesnt need to be a chapter.

2.4 Audience

is it too basic / obvious to just say that I'll youtube it live and stream it on my Twitch??

3. OUTCOMES

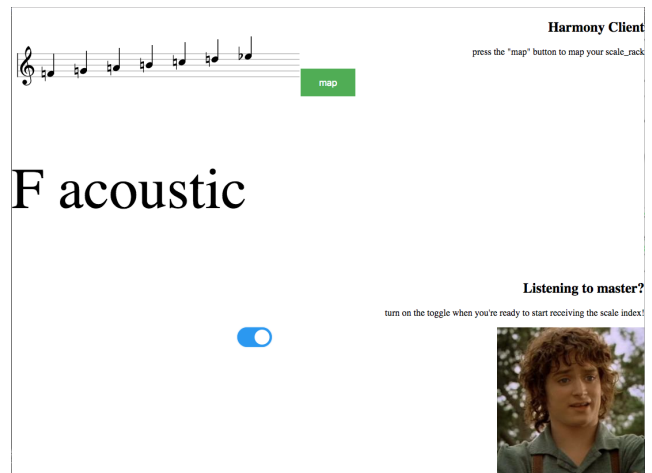


Figure 3: Client page

The Scale Navigator's flexibility means

3.1 Composing

-my own compositional practice with these tools / feedback loop of inspiration -actually composing specific pieces

'Network Impressions' is an electro-acoustic improvisational chorale for SATB, live electronics and interactive score, premiered on December 1st 2018 at CalArts by the Chordas vocal ensemble. The singers explore a series of 16 of these scales, flattening or sharpening a single note with each transition. They are assisted in this by an interactive score that I built in JavaScript, which also triggers electronic and recorded sounds in a concurrently running Ableton session. I chose different texts for each singer at random from the toetoba corpus of english sentences.

Network Impressions

Nathan Turezan

Figure 4: Score for Network Impressions

3.2 Performing

-talk about performing with this tool as a live instrument
-talk about performing the above pieces that I wrote -talk about network music, my choir, wavecave, machines and strings

'INTERSTICES' is an improvisational piece of networked music composed specifically for CalArt's Machine Orchestra, premiered December 6th, 2018. The composition is centered around my Scale Network Navigator, which uses websockets to deliver musical notation and MIDI messages

to an indeterminate number of client web pages open on the laptops of the participating members of the ensemble. The musicians, who sight read the notes, play this music on acoustic instruments. They can also choose to use the incoming MIDI messages to make sound design decisions in Ableton. This way of networking instrumentalists and electronic musicians—and keeping them in the same key—is assisted formally by an animated score that plays on a screen.

‘Modal Intersections’ was an interactive audiovisual installation for the CalArts WaveCave that juxtaposed scenes of roadway intersections in Los Angeles with harmonic intersections in a network of 7-note scales. The Scale Navigator was open on an iPad embedded in a podium in the middle of the space. Users were invited to touch the screen and navigate simultaneously through a series of video loops of Los Angeles and harmonic keys. This was accomplished via MIDI messages sent from the Scale Network Navigator—when triggered by the user—to Ableton, which controlled the music, and TouchDesigner, which was responsible for the visuals. Two opposing DisKlavier player pianos in the room played melodies and arpeggios in whatever scale that was selected, accompanied by a corresponding video loop of an LA intersection: a highway intersection for each harmonic intersection.

3.3 Issues arising from Composition and Performance

-sight reading is hard. The sight-reader’s attention must not be diverted, which can be difficult in an ensemble with a conductor

-rehearsal and preparation important especially when improvising

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4. CONCLUSION

in which i conclude everything

5. ACKNOWLEDGMENTS

thank you to Nathan Ho, Dexter Sheperd, Colin Honigman, Ajay Kapur

APPENDIX

A. HEADINGS IN APPENDICES

A.1 Introduction

A.2 The Body of the Paper

A.2.1 Type Changes and Special Characters

A.2.2 Math Equations

Inline (In-text) Equations.

Display Equations.

A.2.3 Citations

A.2.4 Tables

A.2.5 Figures

A.3 Conclusions

A.4 Acknowledgments

A.5 Additional Authors

A.6 References

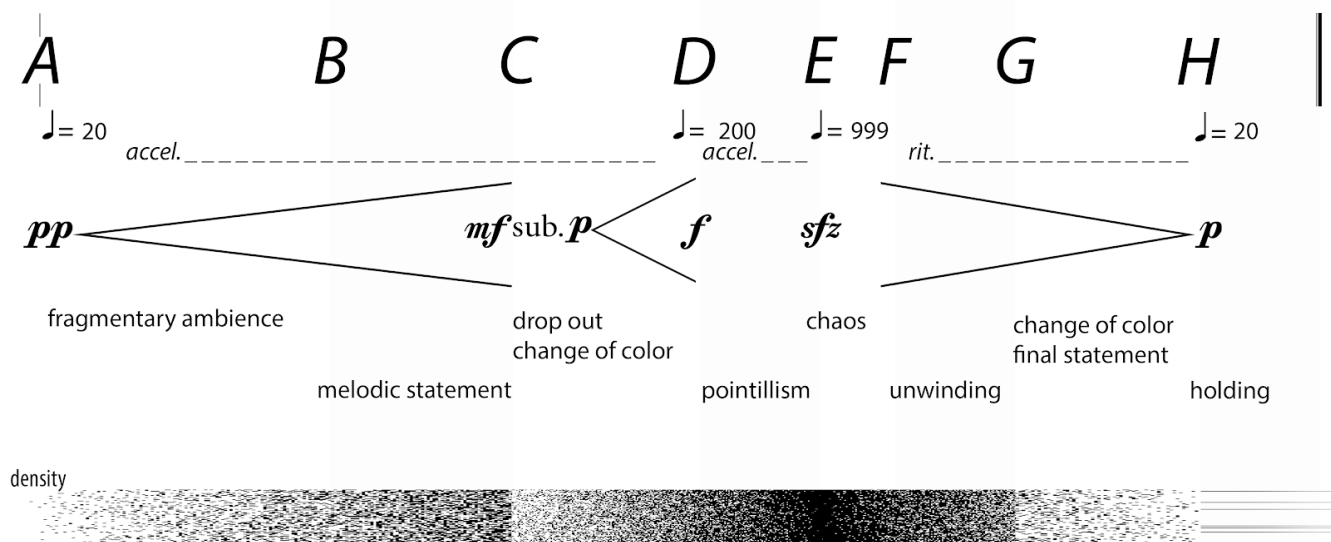


Figure 5: video score for INTERSTICES.